Pulleys and Strings

Answer key

Conventions used throughout

• **Forces:** F_g = gravity ($F_g = mg$), so $F_{ga} = m_a g$ and $F_{gb} = m_b g$. T = tension, F_N = normal force, F_s = static friction (largest value $\mu_s F_N$), F_f = friction on a sliding object ($F_f = \mu F_N$).

• **Strings and pulleys:** a light string over a frictionless pulley has the same tension T all along it. A string that doesn't stretch makes every mass on it move with the same size of acceleration a.

• **Signs (from the lesson):** each mass gets its own + direction, and together they make one +x direction along the string (for example + up on one side of the pulley and + down on the other). Forces are written as sizes, with + or − in the equation for their direction, as the worksheet asks.

• **Direction cases:** when the direction is clear (the heavier side of a frictionless pulley falls, or a hanging mass pulls a block on a table toward the pulley), take +x that way and solve once. Use two cases only when the problem leaves the direction ambiguous (Try These 3): try both choices of +x. If a comes out positive, the system really moves that way; if negative, it moves the other way, or (with friction) that case can't happen.

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• **Final answers:** the scalar with its sign, $a = +1.09$ m/s^2, then the vector, $\vec{a} = 1.09$ m/s^2 [positive x], where +x is the chosen direction along the string.

• $g = 9.81$ m/s^2. Every step keeps full precision; only final answers are rounded.

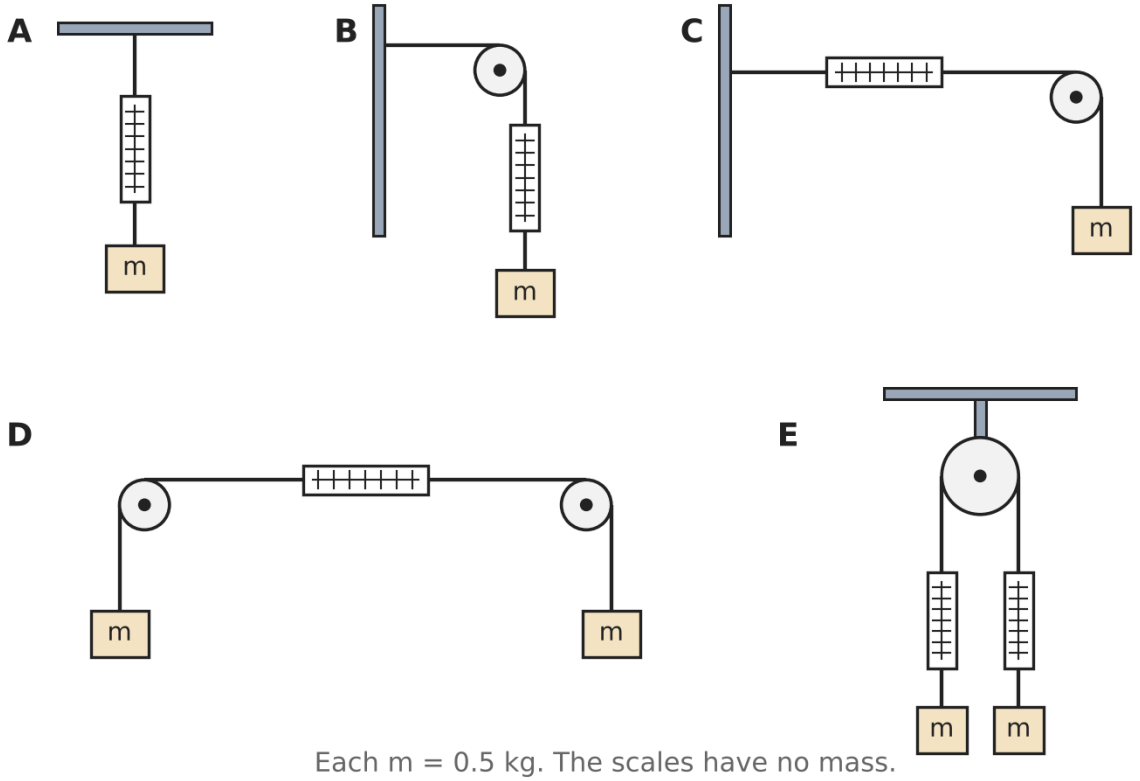
TENSION & PULLEYS

Spring scales in five set-ups

Suppose you were to hang a mass of 0.5 kg in the various configurations shown below. You may assume the scales have no mass.

1. Draw a free body diagram for the spring scale itself in each scenario.

The five set-ups (as on the worksheet)



Each m = 0.5 kg. The scales have no mass.

Step 1 – How hard each string pulls

Each hanging mass is at rest, so the string holding it pulls up exactly as hard as gravity pulls down:

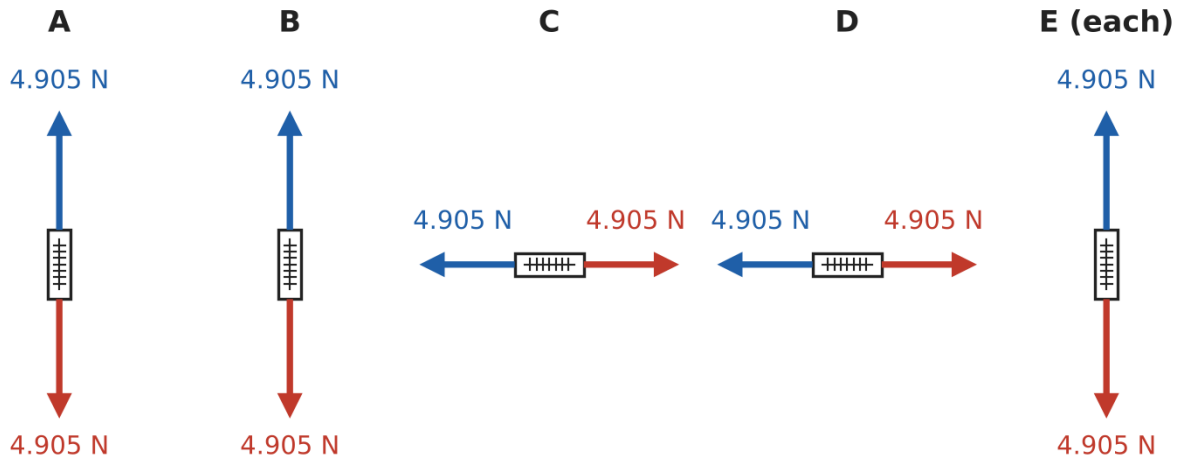
Formula: $F_T = F_g = mg$

$F_T = (0.5\ \text{kg})(9.81\ \text{m/s}^2) = 4.905\ \text{N}$

Step 2 – The forces on each scale

A light string over a frictionless pulley has the same tension all along it, so every string tied to a scale pulls with 4.905 N. The scale has no mass, so no gravity acts on it. It doesn't accelerate, so the pull at one end must be equal and opposite to the pull at the other end:

FBD of the spring scale in each set-up



Every arrow is a tension force $\vec{F}_T$ from a string attached to the scale.

Set-up	What pulls on the two ends of the scale
A	The string to the ceiling pulls up; the mass's string pulls down.
B	The string over the pulley (tied to the wall) pulls up; the mass's string pulls down.
C	The string tied to the wall pulls left; the string over the pulley to the mass pulls right.
D	Each end has a string over a pulley to a hanging mass. One mass does the job the wall did in C, so the scale still feels 4.905 N at each end, not double.
E	On each scale, the string over the pulley pulls up and that scale's mass pulls down.

FINAL ANSWER

Every scale has two equal and opposite tension forces of 4.905 N, one at each end

So every scale reads mg = 4.905 N ≈ 5 N, in all five set-ups (even D)

ATWOOD MACHINE

Equal and unequal masses

Typical pulley problems will consist of an Atwood Machine.

Atwood Machine – consists of two weights tied together and suspended over a pulley as the photo to the right depicts.

1) If the masses are equal, if we were to give one a gentle push to a mass and release it, what type of motion will occur **after** it is released?

2) Suppose that m_b is greater than m_a. Will mass A accelerate? If so, in what direction? Will mass B accelerate? If so, in what direction? How will those accelerations compare in magnitude? Explain.

1) If the masses are equal, if we were to give one a gentle push to a mass and release it, what type of motion will occur **after** it is released?

With equal masses, gravity pulls each side down equally hard, so after the push there is no net force on the system:

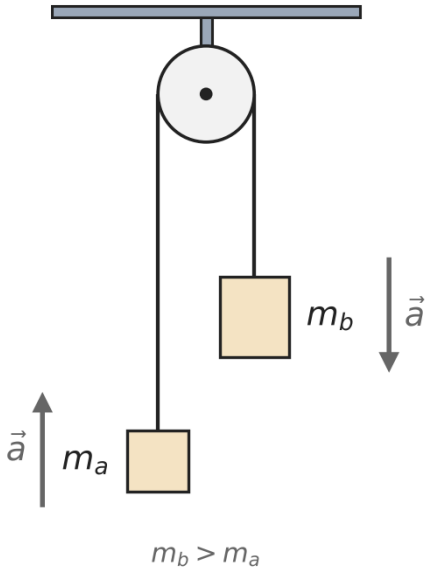
$$F_{net} = F_{gb} - F_{ga} = mg - mg = 0 \quad \Rightarrow \quad a = 0$$

By Newton's first law, the masses keep the velocity the push gave them. Taking +x along the string in the direction of the push, both masses move in the +x direction at the same constant speed (ignoring friction in the pulley).

2) Suppose that m_b is greater than m_a. Will mass A accelerate? ... How will those accelerations compare in magnitude? Explain.

Yes, both accelerate. B is heavier, so gravity pulls B's side down harder than A's side: B speeds up downward and pulls A upward. Following the lesson, choose + for each mass along its own acceleration: + up for A and + down for B. Together these make one +x direction along the string. The string doesn't stretch, so both masses always move the same distance along the string in the same time. Their accelerations have the **same magnitude** and the **same direction, +x**, so one positive a describes both.

The Atwood machine with $m_b > m_a$



FINAL ANSWER

1) Uniform motion: each mass keeps a constant velocity (a = 0)

2) Yes, both accelerate in the +x direction along the string (+ up for A, + down for B), with the same magnitude because the string doesn't stretch

FBDs and symbols

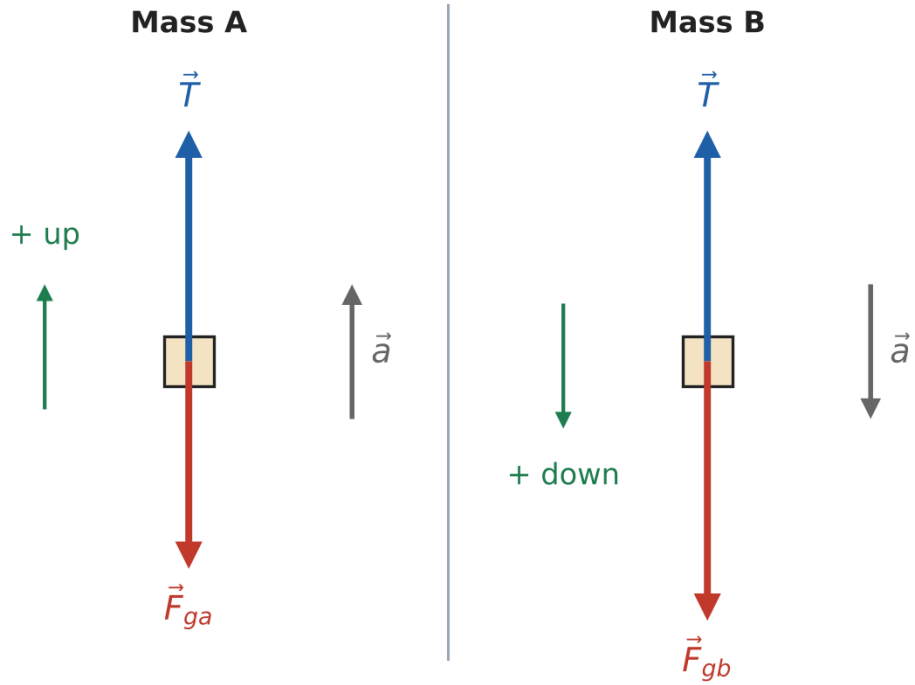
3) Draw a FBD for mass A and for mass B.($m_b > m_a$). Indicate the acceleration of each mass and **choose your sign conventions so they align with the acceleration of the mass**! (positive will be different for each mass)

4) For the FBDs, do you need to use a different symbol for the magnitude of the acceleration of mass A and mass B (i.e. a_a and a_b?) What about for the forces of tension and gravity acting on each mass? Explain.

3) Draw a FBD for mass A and for mass B. Indicate the acceleration of each mass and choose your sign conventions so they align with the acceleration of the mass!

Two forces act on each mass: tension up and gravity down. + is up for A and down for B, so each mass accelerates in its own + direction: both accelerations point along +x. For A, T is longer than F_{ga}; for B, F_{gb} is longer than T. (Drawn to scale for the masses in 5d.)

FBDs of mass A and mass B



4) Do you need to use a different symbol for the magnitude of the acceleration of mass A and mass B? What about for the forces of tension and gravity acting on each mass? Explain.

Quantity	Same or different?	Why
Acceleration	Same symbol, a	The string doesn't stretch, so both sizes are equal. With + along each mass's own motion, a is positive for both.
Tension	Same symbol, T	It is one light string over a frictionless pulley, so it pulls up on A and on B equally hard.
Gravity	Different: F_{ga} and F_{gb}	F_g = mg and the masses are different, so the two weights are different.

FINAL ANSWER

3) A: T up, F_{ga} down, + up. B: T up, F_{gb} down, + down (FBDs above)

4) One a and one T for both masses; gravity needs two symbols, F_{ga} and F_{gb}

ATWOOD MACHINE

The system approach

5) Now we must realize that both masses accelerate **together, as a system**.

a. What force(s) act to move the system forward?

b. What force(s) act to oppose the motion of the system?

c. Write an scalar equation for the net force on the system:

d. If m_a = 0.200 kg and m_b = 0.250 kg, determine the acceleration of the system:

Picture the string pulled straight: the whole system then moves along one line, +x. "Forward" means the +x direction. This question says $m_b > m_a$, so take +x as B down and A up.

a. What force(s) act to move the system forward?

Gravity on B, F_{gb}. It pulls B down, which is +x for B.

b. What force(s) act to oppose the motion of the system?

Gravity on A, F_{ga}. It pulls A down, but +x for A is up. (Tension is inside the system: it pulls A forward and B backward by the same amount, so it cancels.)

c. Write an scalar equation for the net force on the system:

$$F_{net} = F_{gb} - F_{ga}$$

d. If m_a = 0.200 kg and m_b = 0.250 kg, determine the acceleration of the system:

GIVEN

$m_a = 0.200 \text{ kg}$ $m_b = 0.250 \text{ kg}$ $a = ?$

+x is B down, A up

F_{gb} points along +x and F_{ga} points against it. Solve for a in letters first:

Formula: $F_{net} = ma$

Put in the forces and the total mass: $m_b g - m_a g = (m_a + m_b)\, a$

Factor out g: $(m_b - m_a)\, g = (m_a + m_b)\, a$

Divide both sides by $(m_a + m_b)$: $a = \dfrac{(m_b - m_a)\, g}{m_a + m_b}$

$$a = \frac{(0.250 - 0.200)(9.81)}{0.200 + 0.250} = \frac{0.4905}{0.450} = +1.090 \text{ m/s}^2$$

a is positive, so both masses really accelerate in the +x direction (+x is down for B and up for A).

FINAL ANSWER

a. F_{gb} (gravity on mass B)

b. F_{ga} (gravity on mass A)

c. $F_{net} = F_{gb} - F_{ga}$

d. $a = +1.09$ m/s^2

$\vec{a} = 1.09$ m/s^2 [positive x], the same for both masses

ALTERNATIVE METHOD

Calculate acceleration with two equations

1. Go back to your **FBD** of **mass A**. Use **Newton's 2nd Law** to write down an equation that relates the forces acting on **mass A** to the **acceleration** of **mass A**. Write the forces as scalars and use + or – to indicate direction. (Don't solve for *a*.)
2. Use **Newton's 2nd Law** to write down a **scalar equation** that relates the forces acting on **mass B** to the acceleration of **mass B**.
3. Algebraically eliminate ***T*** from the above two equations and solve for the acceleration of the masses.
4. Use an equation from question **8**. or **9**. to solve for the magnitude of the force of tension. How does this value compare with F_{ga} and F_{gb}? Explain why.

Use the +x directions from the FBDs: + up for A and + down for B.

1. Newton's 2nd Law for mass A (+x up)

T points up (+) and F_{ga} points down (−):

Formula: $F_{net} = ma$

Mass A: $T - m_a g = m_a a$

2. Newton's 2nd Law for mass B (+x down)

F_{gb} points down (+) and T points up (−):

Mass B: $m_b g - T = m_b a$

3. Algebraically eliminate T from the above two equations and solve for the acceleration of the masses.

T is added in one equation and subtracted in the other, so adding them cancels T. Solve in letters, then put in the numbers:

Add the equations (the T's cancel): $m_b g - m_a g = m_a a + m_b a$

Factor both sides: $(m_b - m_a)\, g = (m_a + m_b)\, a$

Divide both sides by $(m_a + m_b)$: $a = \dfrac{(m_b - m_a)\, g}{m_a + m_b}$

$$a = \frac{(0.250 - 0.200)(9.81)}{0.450} = +1.090 \text{ m/s}^2$$

The same formula and answer as the system method in 5d.

4. Use an equation from question 8. or 9. to solve for the magnitude of the force of tension. How does this value compare with F_{ga} and F_{gb}? Explain why.

"Question 8. or 9." means the two equations from 1. and 2. above. Solve mass A's equation for T:

Mass A: $T - m_a g = m_a a$

Add $m_a g$ to both sides and factor: $T = m_a(g + a)$

$T = (0.200)(9.81 + 1.090) = 2.180 \text{ N}$

Compare: F_{ga} = 1.962 N < T = 2.180 N < F_{gb} = 2.4525 N. A speeds up upward, so the upward pull on it (T) must be bigger than its weight. B speeds up downward, so its weight must be bigger than the upward pull (T). T has to be between the two weights.

FINAL ANSWER

1. $T - m_a g = m_a a$

2. $m_b g - T = m_b a$

3. $a = +1.09 \text{ m/s}^2$; $\vec{a} = 1.09 \text{ m/s}^2$ [positive x]

4. T = 2.18 N, which is between F_{ga} (1.96 N) and F_{gb} (2.45 N)

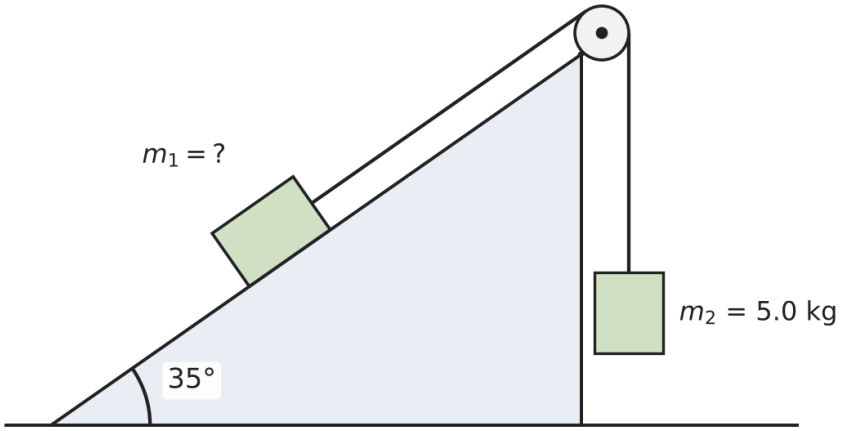
Check: mass B's equation gives the same tension: $T = m_b(g - a) = (0.250)(9.81 - 1.090) = 2.180$ N.

EXAMPLE 1

Largest mass that stays at rest

Example 1) The figure below shows two blocks joined with a rope that runs over a pulley. The mass of $\mathbf{m_2}$ is **5.0kg**, and the incline is **35°**. The coefficient of static friction between $\mathbf{m_1}$ and the inclined plane is **0.25**. Determine the largest mass for $\mathbf{m_1}$ such that both blocks remain at **rest**.

Picture the situation



$\mu_s = 0.25$ between m_1 and the ramp

m_2 pulls m_1 up the ramp through the rope. If m_1 is heavy enough, it slides down the ramp and pulls m_2 up.

GIVEN

$m_2 = 5.0$ kg	$\theta = 35°$	$\mu_s = 0.25$
$a = 0$	$T = ?$	$m_1 = ?$

The question asks for the largest m_1, so the direction is clear: if m_1 were any heavier, it would slide down the ramp and pull m_2 up. At the largest mass the blocks are just about to slip that way, so static friction is at its largest, $F_s = \mu_s F_N$, and points up the ramp, against the slip.

Tension and friction

m_2 is at rest, so the rope holds its weight. On the ramp, the normal force balances the part of m_1's weight into the ramp:

Mass 2, at rest: $T - m_2 g = 0 \quad \Rightarrow \quad T = m_2 g$

Mass 1, y direction: $F_N - m_1 g\cos\theta = 0 \quad \Rightarrow \quad F_N = m_1 g\cos\theta$

Largest static friction: $F_s = \mu_s F_N = \mu_s m_1 g\cos\theta$

m_1 about to slide down the ramp (+x down the ramp for m_1, + up for m_2)

F_{g1x} points along +x. T and F_s point up the ramp (−x):

Formula: $F_{net} = ma \quad (a = 0)$

Mass 1: $m_1 g\sin\theta - T - \mu_s m_1 g\cos\theta = 0$

Put in $T = m_2 g$: $m_1 g\sin\theta - \mu_s m_1 g\cos\theta = m_2 g$

Divide by g, factor out m_1: $m_1(\sin\theta - \mu_s\cos\theta) = m_2$

Divide both sides by $(\sin\theta - \mu_s\cos\theta)$: $m_1 = \dfrac{m_2}{\sin\theta - \mu_s\cos\theta}$

$$m_1 = \frac{5.0}{\sin 35° - (0.25)\cos 35°} = \frac{5.0}{0.3688} = 13.56 \text{ kg}$$

Any heavier m_1 would need more friction than the ramp can give, so m_1 would slide down. This is the largest mass.

FINAL ANSWER

Largest m_1 = 14 kg

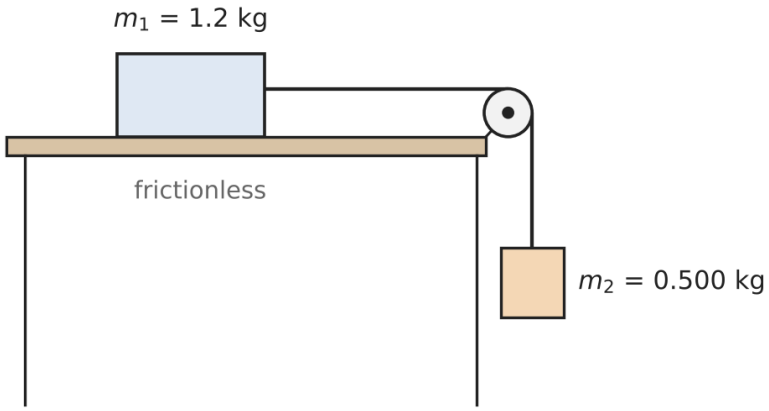
Check: with m_1 = 13.56 kg: $m_1 g \sin\theta$ = 76.29 N, and T + F_s = 49.05 + 27.24 = 76.29 N.

TRY THESE 1

Block on a frictionless table

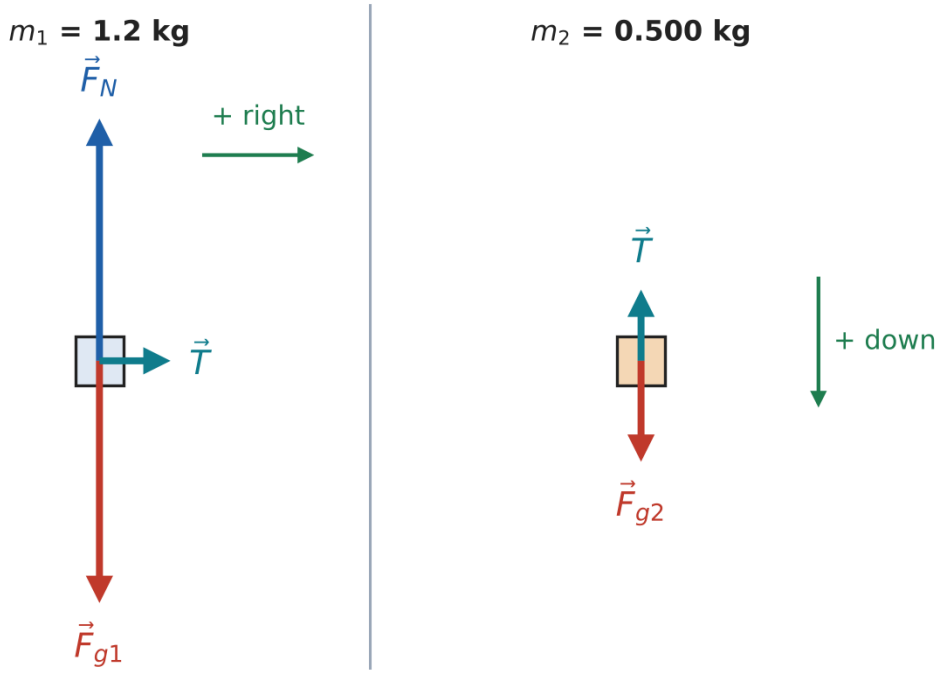
1. In the diagram below, m_1 = 1.2 kg and m_2 = 0.500 kg. The two masses are connected by an inextensible string and m_1 slides along a frictionless surface. Find the acceleration of the system and the string tension. (2.9 m/s^2 3.5 N)

Picture the situation



m_2 hangs from the string; m_1 sits on a frictionless table.

FBDs (+x: m_1 toward the pulley, m_2 down)



On m_1, F_N and F_{g1} cancel and there is no friction. Tension is inside the system, so the only outside force along the string is m_2's weight. That weight is the only thing driving the system, so it must move with m_2 falling: take +x that way.

+x is m_2 down (m_1 toward the pulley)

m_2g points along +x:

Formula: $F_{net} = ma$

Put in the forces and the total mass: $m_2g = (m_1 + m_2)\,a$

Divide both sides by $(m_1 + m_2)$: $a = \dfrac{m_2g}{m_1 + m_2}$

$$a = \frac{(0.500)(9.81)}{1.2 + 0.500} = \frac{4.905}{1.7} = +2.885 \text{ m/s}^2$$

Positive, so both masses accelerate in the +x direction.

Tension

Tension is the only sideways force on m_1. Solve in letters, then put in the numbers:

Mass 1: $T = m_1a$

Put in $a = \dfrac{m_2g}{m_1 + m_2}$: $T = \dfrac{m_1m_2g}{m_1 + m_2}$

$$T = \frac{(1.2)(0.500)(9.81)}{1.2 + 0.500} = 3.462 \text{ N}$$

FINAL ANSWER

a = +2.9 m/s^2

$\vec{a}$ = 2.9 m/s^2 [positive x], the same for both masses

T = 3.5 N

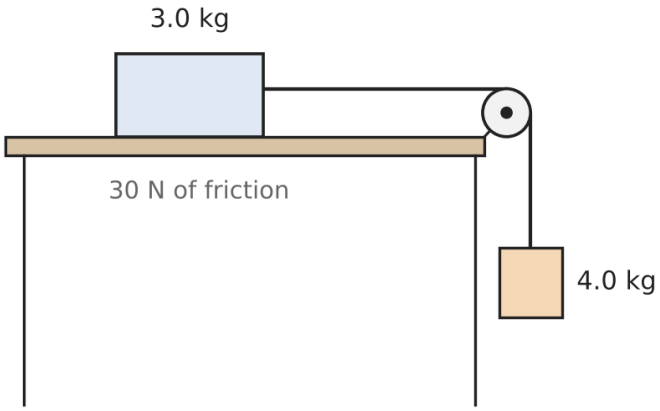
Check: mass 2: $m_2g - T = 4.905 - 3.462 = 1.443$ N $= (0.500)(2.885)$.

TRY THESE 2

Tension without finding the acceleration

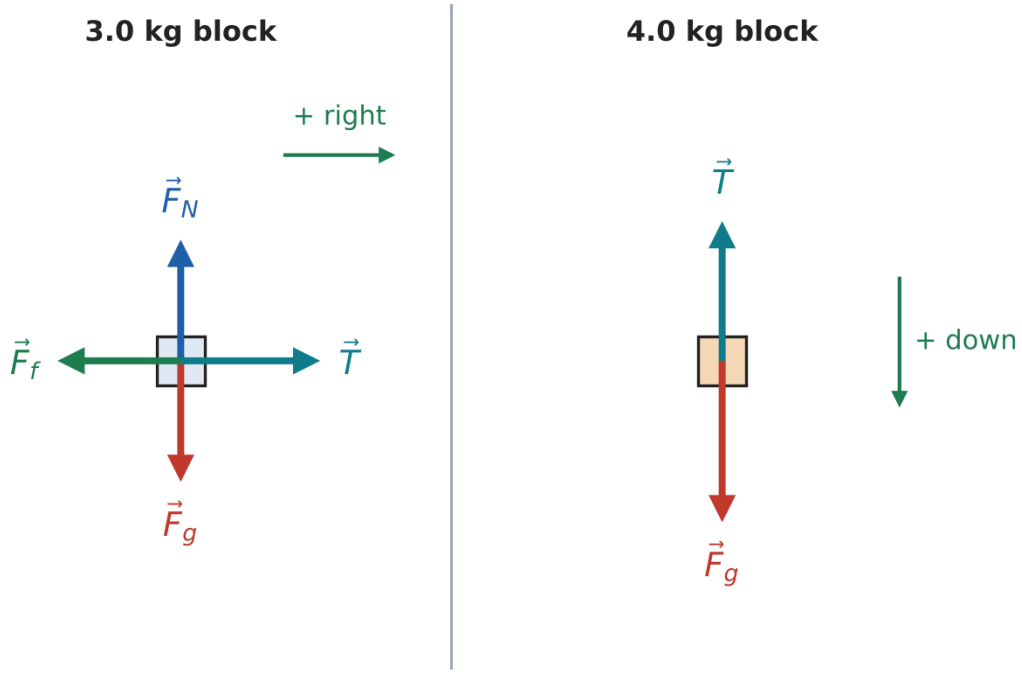
2. Find the tension in the string connecting the two blocks. Do not find the acceleration. There is 30 N of friction between the block and the table. The string is light and the pulley is frictionless. [34 N]

Picture the situation



The 3.0 kg block is on the table with 30 N of friction; the 4.0 kg block hangs.

FBDs (+x: 3.0 kg toward the pulley, 4.0 kg down)



Call the masses m_1 = 3.0 kg (table) and m_2 = 4.0 kg (hanging), and F_f = 30 N. The hanging block's weight is the only force that can drive the system, so m_2 falls and m_1 slides toward the pulley: take +x that way. Friction opposes the motion, so it points away from the pulley.

+x is m_2 down (m_1 toward the pulley)

Friction on m_1 points away from the pulley (−x). One equation for each block:

Formula: $F_{net} = ma$

m_1 (+x toward the pulley): $T - F_f = m_1 a$

m_2 (+x down): $m_2 g - T = m_2 a$

Solve each for a and set them equal (same a), then isolate T in letters:

Divide each by its mass: $\dfrac{T - F_f}{m_1} = \dfrac{m_2 g - T}{m_2}$

Multiply both sides by $m_1 m_2$: $m_2(T - F_f) = m_1(m_2 g - T)$

Expand: $m_2 T - m_2 F_f = m_1 m_2 g - m_1 T$

Add $m_1 T$ and $m_2 F_f$ to both sides: $(m_1 + m_2)\,T = m_1 m_2 g + m_2 F_f$

Divide both sides by $(m_1 + m_2)$: $T = \dfrac{m_2(m_1 g + F_f)}{m_1 + m_2}$

$$T = \frac{(4.0)[(3.0)(9.81) + 30]}{3.0 + 4.0} = \frac{(4.0)(59.43)}{7.0} = 33.96 \text{ N}$$

FINAL ANSWER

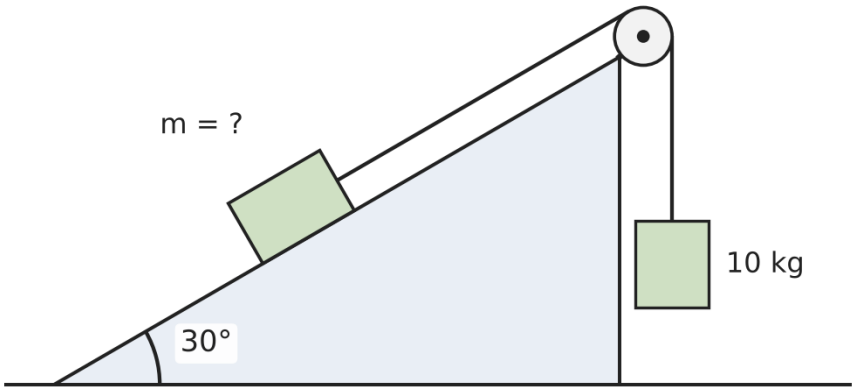
T = 34 N

TRY THESE 3

Finding the mass on the ramp

3. What is the value of *m* that would give the system an acceleration of 2.0 m/s/s? The coefficient of friction is 0.15, the string is light and the pulley is frictionless. [9.5 kg]

Picture the situation



$\mu = 0.15,\ \vec{a} = 2.0\ \text{m/s}^2$

From the worksheet's diagram: a 30° ramp, with a 10 kg mass hanging from the rope.

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$a = 2.0\ \text{m/s}^2$	$\mu = 0.15$	$\theta = 30°$
$M = 10\ \text{kg}$	$T = ?$	$m = ?$

Here the direction is ambiguous: the worksheet gives a but not which way the system moves, and m is unknown, so either side could be the heavier pull. Try both cases. On the ramp, $F_N = mg\cos\theta$ in both cases, so $F_f = \mu mg\cos\theta$, pointing against the motion.

Case 1 – +x is M down (m up the ramp)

Friction on m points down the ramp (−x).

Formula: $F_{net} = ma$

M (+x down): $Mg - T = Ma \quad \Rightarrow \quad T = M(g - a)$

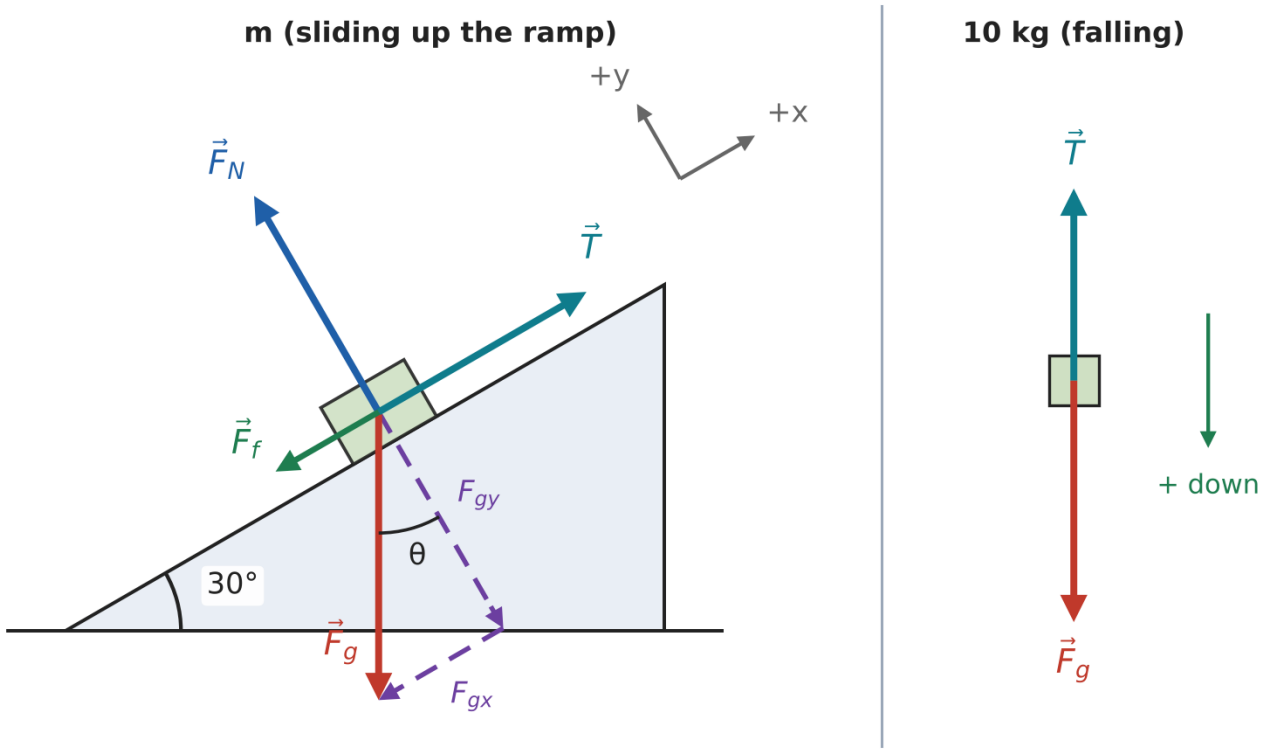
m (+x up the ramp): $T - mg\sin\theta - \mu mg\cos\theta = ma$

Put in T and move the m terms together:

$$M(g - a) = m(a + g\sin\theta + \mu g\cos\theta)$$

Divide: $$m = \frac{M(g - a)}{a + g\sin\theta + \mu g\cos\theta}$$

FBDs for Case 1 (m: +x up the ramp; 10 kg: +x down)



Put in the numbers:

$$m = \frac{(10)(9.81 - 2.0)}{2.0 + (9.81)\sin 30° + (0.15)(9.81)\cos 30°}$$

$$m = \frac{78.10}{8.1794} = 9.548 \text{ kg}$$

Case 2 – +x is M up (m down the ramp)

Now friction on m points up the ramp (−x for m).

M (+x up): $T - Mg = Ma \quad \Rightarrow \quad T = M(g + a)$

m (+x down the ramp): $mg\sin\theta - T - \mu mg\cos\theta = ma$

Put in T and solve the same way: $m = \dfrac{M(g + a)}{g\sin\theta - \mu g\cos\theta - a}$

$$m = \frac{(10)(9.81 + 2.0)}{(9.81)\sin 30° - (0.15)(9.81)\cos 30° - 2.0}$$

$$m = \frac{118.1}{1.6306} = 72.4 \text{ kg}$$

This case also works: a 72 kg block sliding down the ramp would pull the 10 kg mass up with the same 2.0 m/s^2. The worksheet's answer, 9.5 kg, is Case 1.

FINAL ANSWER

Case 1 (10 kg falls): m = 9.5 kg — the worksheet's answer

Case 2 (10 kg rises): m = 72 kg also gives a = 2.0 m/s^2

Check: Case 1: T = 78.10 N, and T − mg sin 30° − μmg cos 30° = 19.10 N = (9.548)(2.0).